

SeokHyeon (Lucas) Kong

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RESEARCH INTERESTS

Quantum Computer Systems—Architecture, Software Stack, Error Correction and Detection

EDUCATION

Sungkyunkwan University (SKKU)

Bachelor of Science in Electronic and Electrical Engineering
Bachelor of Science in Advanced Semiconductor Engineering

- **Cumulative GPA: 3.95 / 4.0 | Major GPA: 4.0 / 4.0**

Seoul, South Korea

Expected February 2026
System Architecture Track

The Pennsylvania State University

Exchange Student in Computer Engineering

- **Grade: 4.0 / 4.0**

University Park, PA

Fall 2024

PUBLICATIONS & PRESENTATION

S. H. Kong, D. Kim, K. Maeng*, and E. Seo*, “Characterizing the System Overhead of Discrete Gaussian Noise Generation for Differential Privacy,” *IEEE Computer Architecture Letters*, under review, 2025

S. H. Kong, C. Park, J. Moon, and D. Dahl*, “Solving Constraint Satisfaction Problems with Variational Quantum Algorithm on NISQ Devices,” *Poster presented at the Quantum Information Research Support Center (Qcenter)*, Suwon, South Korea, Sep. 2025

S. H. Kong and H. Oh*, “Predicting Key Regional Real Estate Prices Using Machine Learning Technique: with an emphasis on Jeonsae system,” *Journal of the Korean Institute of Information and Communication Engineering*, vol. 27, no. 10, pp. 1208–1213, Oct. 2023

RESEARCH EXPERIENCE

IonQ

Research Mentee, Advised by Dr. Denny Dahl

Remote

08/2025–09/2025

- Addressed the Instant Insanity problem on NISQ devices by employing a graph-theoretic encoding and a divide-and-conquer approach with two distinct post-processing methods.
- Applied a Variational Quantum Algorithm to the Four Corners Map Coloring problem.

The Pennsylvania State University

Undergraduate Researcher, Advised by Prof. Kiwan Maeng

University Park, PA

08/2024–09/2025

- Conducted in-depth GPU profiling with Nsight Compute and Nsight Systems to analyze performance bottlenecks and system-level overhead in differentially private training.
- Proposed an optimization that resolved the dominant Geometric operation overhead, reducing it from 50% to 19% while preserving Gaussian noise quality.

Arch Lab at SKKU**Suwon, South Korea***Undergraduate Research Intern, Advised by Prof. Dongmoon Min**07/2025–08/2025*

- Built foundational expertise in Quantum / Cryogenic Computer Systems through literature reviews, technical presentations and academic defenses.
- Investigated quantum computer systems, including superconducting-qubit design, error mechanisms, control and readout techniques as well as cryogenic computer architectures.

POSTECH QCQN Lab**Pohang, South Korea***Undergraduate Research Intern**01/2025*

- Performed 397 nm fluorescence (Doppler) and 729 nm quadrupole spectroscopy on Ca^+ ions and enhanced system stability through targeted adjustments.
- Implemented micromotion optimization via 2D scanning with DC compensation rods.

REFERENCES

Kiwan Maeng, Ph.D. (Advisor)*Assistant Professor, Computer Science and Engineering, The Pennsylvania State University*

- Email: kvm6242@psu.edu

Denny Dahl, Ph.D. (Advisor)*Customer Success Director, IonQ*

- Email: denny.dahl@ionq.co

HONORS & AWARDS

Qiskit Global Summer School – Quantum Excellence (IBM Quantum) *2025*Quantum Hackathon – Director’s Award *2025*President’s List (2 consecutive years) *2024, 2025*Dean’s List (5 consecutive semesters) *SP2023, FA2023, SP2024, FA2024, SP2025*Korea-U.S. Student Exchange Scholarship in the field of high-tech industry (\$9,000) *FA2024***TEACHING EXPERIENCE**

Academic Tutor, Sungkyunkwan University (SKKU)**Suwon, South Korea***Intro to Electromagnetism**Fall 2023***Teaching Assistant**, Sungkyunkwan University (SKKU)**Seoul, South Korea***Engineering Mathematics 2**Summer 2023***TECHNICAL SKILLS**

Software: Qiskit, CUDA, C, Python, MATLAB, LaTeX**Hardware:** System Verilog, Verilog, Verdi, Virtuoso, Nsight Systems, Nsight Compute**LEADERSHIP & SERVICE**

Executive Board, President’s List (Honor Society at SKKU)*2025***Qiskit Advocate**, IBM Quantum*2025***Squad Leader**, Republic of Korea Army (Military Service)*04/2021–10/2022*